

Computational Validation and Generated Graphics

To verify the internal consistency of the 6D Unified Field Theory and to illustrate its phenomenological consequences, we implemented a numerical computational package ('MVHristov_6D.UFT.Computations.py'). The code successfully generates four classes of plots corresponding to the main testable predictions of the theory:

1. **Galaxy Rotation Curves:** Using the Yukawa-corrected gravitational potential derived in the 6D framework, the rotation velocities of Milky Way-scale galaxies were computed. The 6D prediction smoothly reduces to Newtonian gravity when the correction parameters are small, confirming internal consistency. The resulting curves reproduce the characteristic flattening pattern without requiring particle dark matter.
2. **Decoherence Scaling:** The geometric decoherence rate $\Gamma \propto (M/m_p)^{2/3}$ predicts a monotonic decrease of coherence time with increasing mass. The numerical plot spans more than 25 orders of magnitude and shows a smooth power-law behavior consistent with the analytical expression.
3. **Short-Range Gravity Deviations:** For sub-millimeter distances the theory predicts Yukawa-type corrections to the Newtonian force. The generated curves show that Newtonian gravity is recovered at large separations, while measurable deviations appear in the 10^{-4} – 10^{-2} m range, in accordance with the chosen correlation lengths.
4. **Atomic Clock Fluctuations:** Simulated fractional frequency noise arising from internal temporal geometry produces a correlated time-domain signal and a power spectral density with the expected $1/f$ -type falloff. The amplitudes lie within the range testable by next-generation optical clocks.

All computations executed without numerical instability or pathological behavior. Together, these graphics provide a concise demonstration that the 6D framework produces well-defined, testable predictions in astrophysics, macroscopic quantum behavior, precision metrology, and short-range gravity.